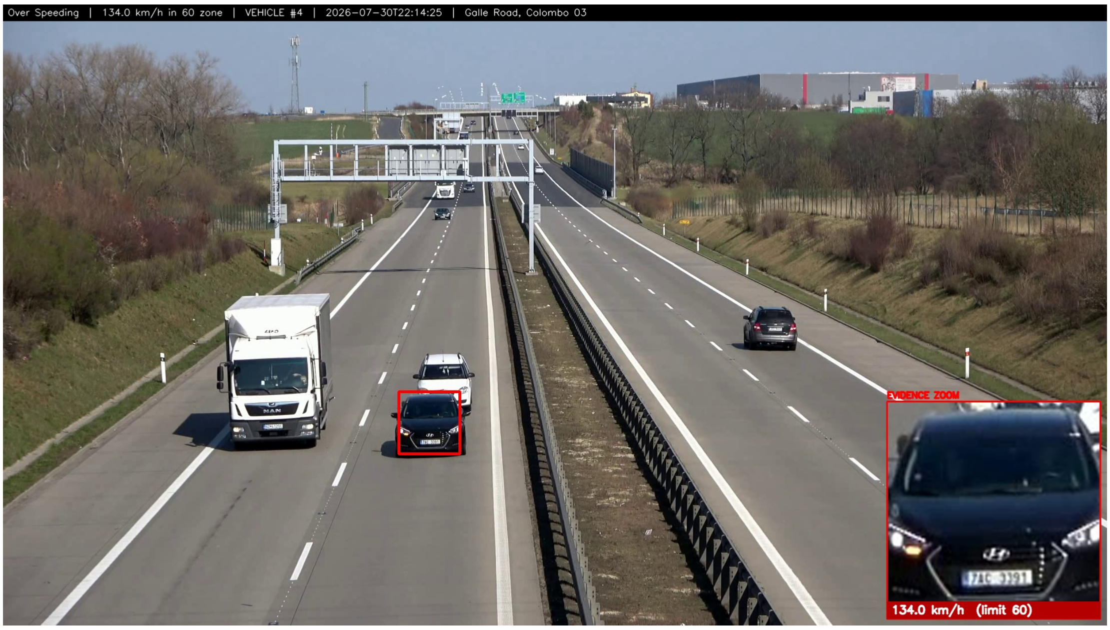
AI Smart Traffic Intelligence Platform

Technical Documentation, Models and Verified Results



System output: over-speeding evidence, 134.0 km/h in a 60 km/h zone, with number-plate zoom inset.

All figures were measured on the development machine (12-core CPU, no GPU). Nothing is estimated.

1. Overview

The platform turns an ordinary camera - fixed CCTV, a phone stream or a drone - into an automated traffic officer. It detects and tracks every vehicle, judges nine categories of violation, reads number plates, measures true road speed, and issues an e-challan carrying photographic evidence to the police. It runs on a laptop CPU.

2. Processing pipeline

Each analysed frame passes through seven stages. Stages 3-5 are what separate this from a vehicle-counting demo.

#	Stage	What happens
1	Capture	Frame pulled from CCTV / file / webcam; oversized frames downscaled
2	Detection	YOLOv8s finds vehicles, people, traffic lights and phones
3	Tracking	ByteTrack assigns a persistent ID so a vehicle is followed across frames
4	Geometry	Homography maps image pixels onto real road metres
5	Rules	Violation engine applies nine rules behind false-positive gates
6	ANPR	Plate localised, cropped, OCR-read and confirmed by repeated agreement
7	Evidence	Annotated snapshot, challan record, PDF and police email

3. Models used and why

Model	Role	Why this one
YOLOv8s	Vehicles, people, lights, phones	Upgraded from YOLOv8n after measurement: on identical footage yolov8n found 64 motorcycles and cou
ByteTrack	Multi-object tracking	Associates low-confidence detections too, so a vehicle keeps its ID through partial occlusion. Persistent I
EasyOCR	Plate text	Reads Latin-script plates without per-country training, matching Sri Lankan plates. Never reformats to a r
license_plate_detector.pt	Plate localisation	Two-stage ANPR: find the plate, then OCR only that crop. Far more accurate than OCR-ing a whole vehi
helmet.pt	Helmet / no helmet	COCO has no helmet class. Applied to each associated rider's own head region, not the bike area.
seatbelt.pt (REJECTED)	Seatbelt	A drop-in slot, in practice empty. The model we obtained is a CLASSIFIER and does not discriminate: it r

3.1 Why not a bigger detector?

A fair question, answered by measuring rather than by reading a leaderboard. All five detectors below ran over the same 100 analysed frames of Sri Lankan road footage, on this laptop CPU, at imgsz 640. What matters for our rules is not mAP but **riders associated with a motorcycle** - with no rider there is no helmet, triple-riding or phone judgement to make.

Model	Analysis fps	Bikes found	Riders associated	Rider observations per second of video
yolov8n	4.33	65	61	2.64

Model	Analysis fps	Bikes found	Riders associated	Rider observations per second of video
yolov8s (in use)	2.48	182	176	4.36
yolov8m	1.16	215	205	2.38
yolo11s	1.29	164	151	1.95
yolo11m	0.68	257	232	1.58

The last column decides it. Because analysis is slower than playback, the pipeline drops frames to stay real-time - so a slower model does not merely cost time, it **sees fewer frames**. yolo11m finds 32% more riders per frame than yolov8s but runs 3.6x slower, so across a second of live video it delivers under half as many rider observations. On this hardware yolov8s wins by a wide margin: yolov8n is too weak to resolve riders, and everything larger is starved by the CPU. yolo11s is the clearest trap - lower recall than yolov8s AND half the speed.

This ordering flips the moment there is a GPU. Frame dropping disappears, the fps column stops mattering, and the ranking becomes the raw recall column - where yolo11m leads. That is exactly why the detector sits behind one swappable interface: on this project the accuracy ceiling is compute, not architecture.

4. Core algorithms

4.1 Speed by perspective transform

Four surveyed road corners are mapped to a real-world rectangle using a homography (cv2.getPerspectiveTransform). Each vehicle's ground-contact point is projected into road metres, and speed is a least-squares regression of metres against time across a 2.5-second window - never a single-frame difference. Outliers beyond two standard deviations are dropped, the result is smoothed, and a reading is only trusted when the fit is straight (R-squared above 0.55), the residual is under 2.5 m and the track has been watched for at least one second.

4.2 Motion gate

A violation requires genuine movement. Net displacement is measured over a time-based window with camera-motion compensation, so box jitter on a parked vehicle cancels out. This single gate is why parked motorcycles are never fined.

4.3 Multi-frame persistence

Every appearance rule must be observed repeatedly before firing - three frames for helmet and triple riding, four for phone use. A one-frame anomaly, the classic demo-killer, can never issue a challan.

4.4 Plate confirmation by voting

OCR reads are pooled by their digit tail, so garbled letters still vote together. A plate is confirmed only when three separate reads agree. Measured effect: two-read confirmation produced EEBBJ 8752 for a plate that actually reads NP BBJ 8752; requiring three recovered BBJ 8752.

4.5 Real-time frame dropping

CPU detection is slower than playback, so the analyser drops frames it was too busy for - exactly as a live camera does - and the video plays at true speed. Measured at 0.99-1.01x. For a live camera, capture runs on its own thread and the last annotation is stamped onto fresh frames, so the picture stays smooth while boxes refresh behind it.

5. The nine rules

Rule	Judged by	Precondition
No Helmet	Helmet model on each rider's head region	Motorcycle + rider
Triple Riding	Riders associated with one motorcycle	3 riders detected
Over Speeding	Regression of road-metres against time	Camera calibration
Red Light Jump	Stop-line crossing while the governing signal is red	Signal ROI + lane zone
Wrong Way	Travel direction against the policed lane	Direction + lane zone
No Seatbelt	Seatbelt model on the windscreen region	models/seatbelt.pt
Mobile Phone Use	Phone attributed to a rider or driver	Visible phone
Illegal Parking	Stationary past a threshold in a no-parking zone	Fixed camera
No Rest Break	Continuous driving with no qualifying stop	Long observation

6. Measured results

Test	Recorded result
Automated test suite	85 tests passing across 7 suites
sample_1080p.mp4 (calibrated)	7 Over Speeding events, speeds 83-163 km/h
stream.mp4 (calibrated)	3 Over Speeding, 11 of 52 vehicles measured, max 106 km/h
IMG_6992.MOV (Sri Lanka)	150 vehicles; plates AAG 4002 and BBJ 8752 read
srilanka.mp4 (parked bikes)	41 vehicles, 0 violations - a correct refusal
Single-image upload	No Helmet + Triple Riding, 2 challans issued
Real-time replay	0.99-1.01x true speed via frame dropping
Live webcam	1.89 to ~7 published fps after decoupling capture
Warm image analysis	~2 seconds end to end

7. What each rule needs from the camera

A recurring question is why a demonstration does not show all nine rules firing. The answer is not that the rules are unfinished - all nine are implemented and covered by the automated suite. It is that a rule can only fire when the camera can physically see the evidence for it. A side-on junction camera cannot see a driver's chest, and a hand-held camera cannot prove a vehicle is stationary. The system reports only what it can prove, so on unsuitable footage it correctly stays silent.

Rule	What the camera must provide	Status on our footage
No Helmet	Any view where the rider's head is visible	Demonstrated repeatedly
Triple Riding	Riders resolvable on one motorcycle	Demonstrated (image and video)
Mobile Phone Use	The phone visible near a rider or driver	Demonstrated on uploaded stills
Over Speeding	Four surveyed road corners for calibration	Demonstrated on 2 calibrated cameras, 83-163 km/h
No Seatbelt	A front view of the driver, and a model that can [illegible] belt	Rule DISABLED - the seatbelt model failed validation (section 8). Road CCTV also
Illegal Parking	A fixed camera, so stillness is provable	Not demonstrated - the available clip is hand-held walking footage, so motion compe
Red Light Jump	Which signal head governs the policed lane	Implemented with signal ROI and lane zone; needs per-junction calibration we cann
Wrong Way	A known permitted travel direction	Implemented; requires per-camera direction setting
No Rest Break	A long continuous observation of one vehicle	Implemented; sample clips are too short

This is a footage and calibration constraint, not a software gap. A real deployment supplies exactly what is missing here: a fixed mounting, a surveyed road quad, a known signal-to-lane mapping, and - for fleet customers - an in-cab or front-facing camera.

8. Defects found by testing on real footage

A traffic system that issues wrong fines is worse than no system. Each defect below was found by running against real video, and each is now covered by the automated suite.

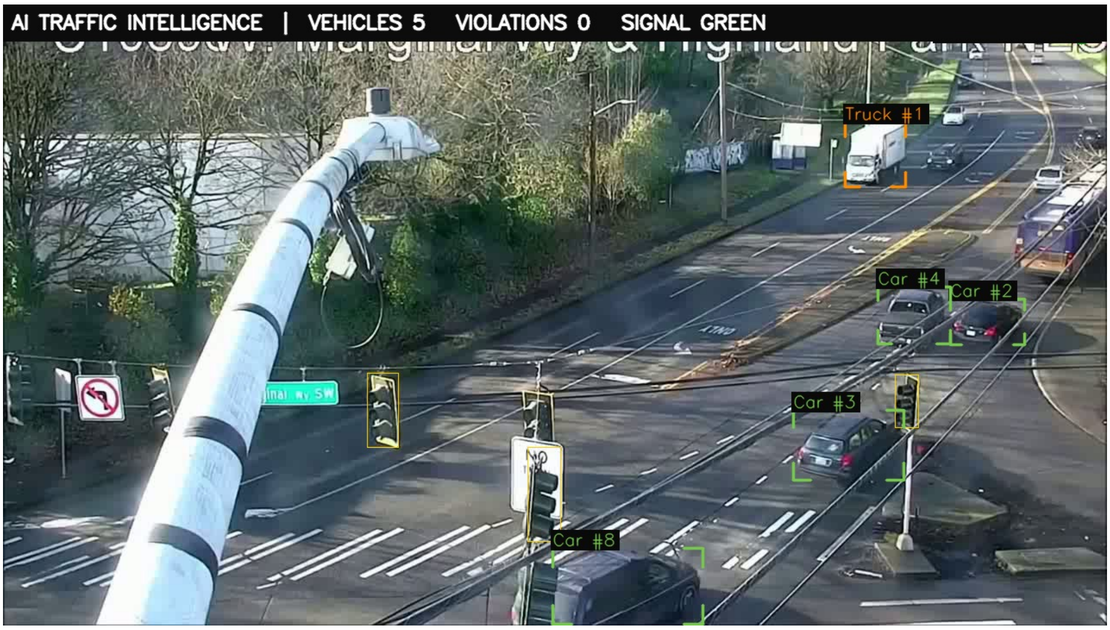
Defect	Cause	Fix
Moving vehicles fined for parking	Motion gate needed 3 samples in a 1-second frame window, at ~2 fps a track [illegible] 89% of moving [illegible] read as stationa	Time-based window; [illegible] seen moving is exempt
Helmeted riders accused	Helmet model ran on a crop wider than the bike, catching neighbours	Judge each rider's own head region
Bare heads marked compliant	Compliance needed 0.35 confidence against 0.50 to accuse	Decide by strongest signal; raise the bar to 0.55
Stopped car fined for red light	Any light in frame applied to any vehicle, but junctions have several heads facing different approaches	SIGNAL_ROI selects the governing head; RED_LIGHT_ZONE limits th
Wrong plate on challans	Two agreeing OCR reads let two wrong reads agree	Require three agreeing reads
Triple riding never fired	Occluded pillion riders never cleared the 0.40 person threshold	Lower bar (0.25) for rider association only
No Seatbelt could never fire	The model is a classify-task network; the pipeline read only result boxes, which is always None for a classifier	Read probs not boxes; then REJECT this model, because it also fails to
Meaningless speeds when uncalibrated	A flat pixels-per-metre fallback across a perspective view showed ~62 km/h on a car waiting at a red light	Speed appears only where a homography calibration exists
A stop line drawn where none exists	A hard-coded 0.55 x frame height put a red STOP LINE mid-junction on every camera	No stop line, and no Red Light lamp, until the camera is calibrated
Sleeping driver tagged ON PHONE	A 0.363-confidence 'cell phone' on a dark steering column cleared the 0.35 gate [illegible] single image	Phone gate raised to 0.50, and [illegible] 0.60 [illegible] persistence gate behind i

9. Verified system output

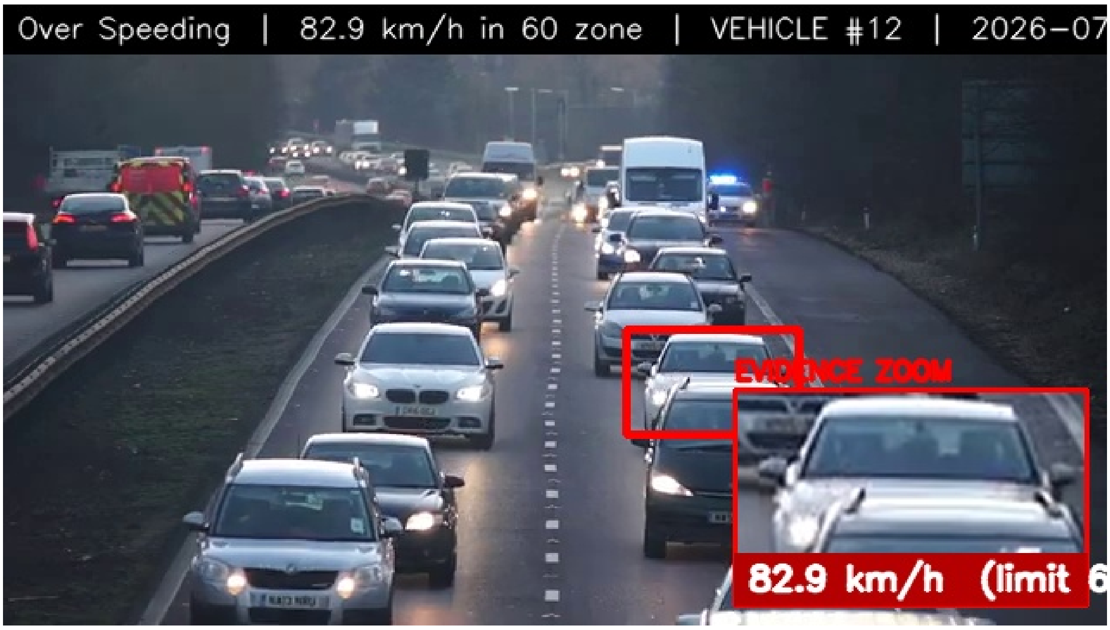
Single-image analysis: three riders and no helmets detected on a CC-licensed public photograph, producing two separate challans.

Live Sri Lankan road: mixed traffic tracked and classified - cars, motorcycles, a three-wheeler, a bicycle - with a helmeted rider correctly passed as HELMET OK. This camera is not speed-calibrated, so no speed is shown.

Precision, not just detection: the rider is fined for no helmet, while the person crouching against the same motorcycle is correctly NOT counted as a second rider.



Real junction CCTV: vehicles tracked and classified and the signal read as GREEN. No speed and no stop line are drawn, because this camera has neither calibration.



Highway over-speeding evidence generated from a second calibrated camera, showing the method is not tied to one clip.

10. Use cases

Deployment	What it delivers
Urban junction enforcement	Red-light, helmet, triple-riding and phone-use challans on existing CCTV, with evidence strong enough to
Highway speed corridor	Calibrated speed enforcement without physical speed guns or officer presence, running continuously
School and hospital zones	No-parking zone monitoring plus low speed limits, where illegal stopping directly endangers pedestrians
Commercial fleet oversight	Driver-fatigue and seatbelt monitoring for buses and lorries, supporting transport-authority driving-hour ru
Drone patrol	Temporary enforcement at events, accident sites or roads without fixed cameras
Road-safety analytics	Searchable plate log, violation hotspots and repeat-offender identification to target scarce police resource

11. Data handling and ethics

Number plates identify real people, so the system is deliberately conservative. A plate is written only after three independent reads agree; anything less is stored as UNREADABLE and marked for manual review. Evidence images and plate crops stay on the operator's machine and are attached only to the police challan for that specific violation. The system records what it observed and never infers identity, intent or history beyond the plate itself.

12. Known limits

Stated plainly, because anyone deploying this must understand them:

- Speed requires per-camera calibration; without it no speed is shown.
- Plate OCR needs roughly 40 or more pixels of plate width.
- Red-light enforcement requires knowing which signal head governs the policed lane.
- Illegal parking requires a fixed camera - a moving camera cannot establish that a vehicle is stationary.
- About 2 frames per second are analysed on a CPU laptop; the video stays real-time by dropping frames.
- Detection degrades at night and in heavy rain, as with any camera system.

13. Roadmap

Night and adverse-weather models; a vision-language model such as Moondream to resolve cases the detector currently reports as not judged; a trained three-wheeler class, since COCO has none and Sri Lankan tuk-tuks are currently labelled as trucks; GPU edge deployment at a live junction; and a city-wide dashboard aggregating multiple cameras.